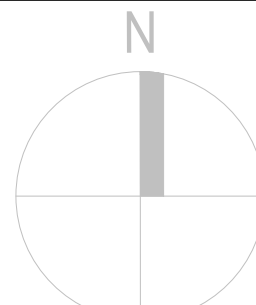
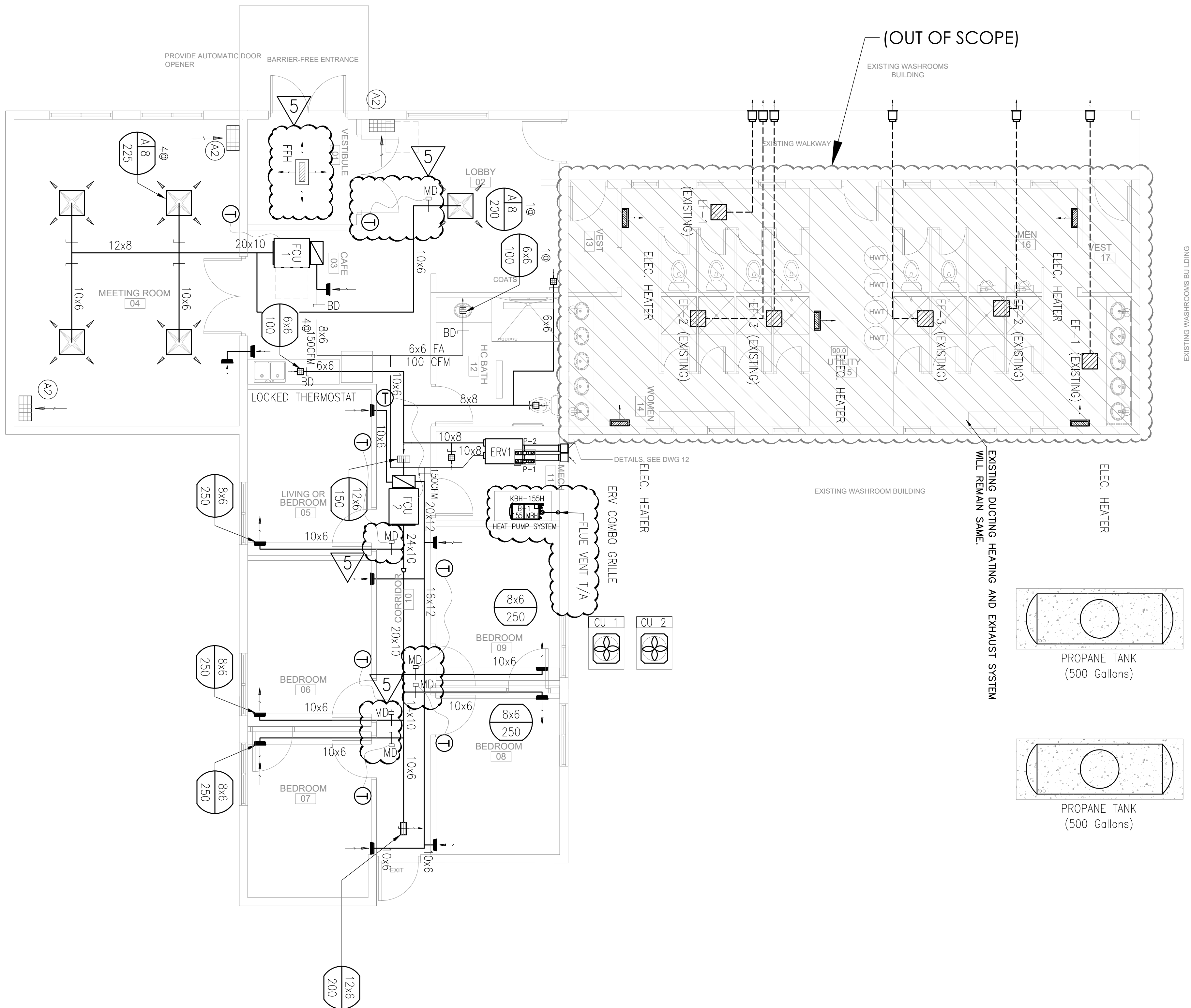




22-174



PROJECT NORTH



1 FLOOR PLAN - 5 DOUBLE BEDROOMS
SCALE: 1/4" = 1' - 0"

NO.	REVISIONS	DATE	BY
5.	DRAWING UPDATED	2026/10/07	M.S.
4.	UPDATED AS PER LATEST ARCH.	2025/09/09	M.S.
3.	UPDATED AS PER LATEST ARCH.	2025/09/03	M.S.
2.	ISSUED FOR TENDER	2023/12/15	M.S.
1.	ISSUED FOR PERMIT	2023/11/20	M.S.

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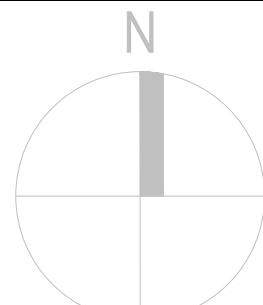
FLOOR PLANS : HVAC LAYOUT PLAN

SCALE : 3/16" = 1' - 0"
PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE
LOCATION : KOC DORM BLDG.

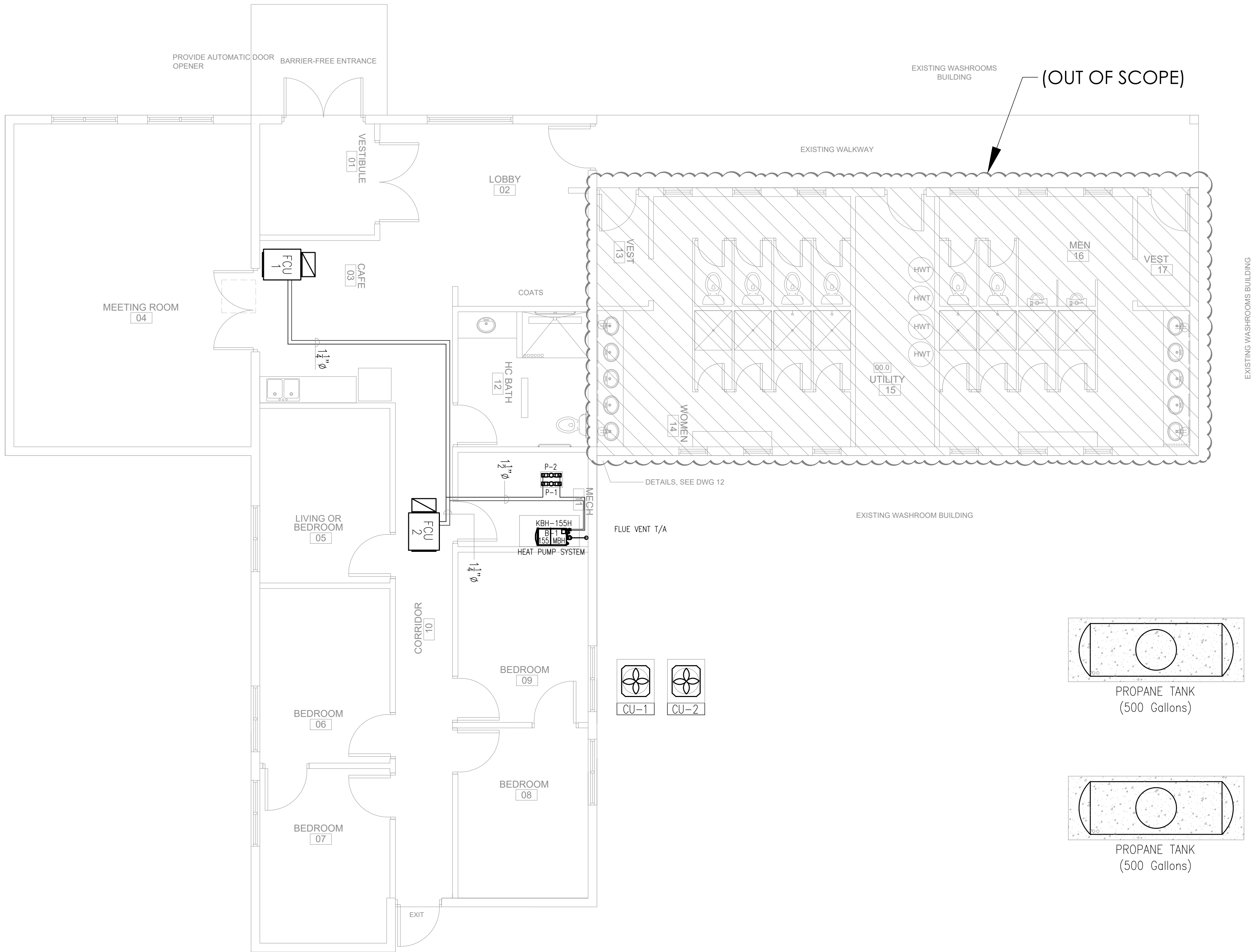
DRAWN BY	M.S.B.	PROJECT NO.	22-174
CHECKED BY	M.N.	DWG. NO.	M2.1
DATE	AUGUST 2023		
ISSUED	AUGUST 2023		



22-174



PROJECT NORTH



1 FLOOR PLAN - 5 DOUBLE BEDROOMS
SCALE: 1/4" = 1' - 0"

NO.	REVISIONS	DATE	BY
5.	DRAWING UPDATED	2026/10/07	M.S.
4.	UPDATED AS PER LATEST ARCH.	2025/09/09	M.S.
3.	UPDATED AS PER LATEST ARCH.	2025/09/03	M.S.
2.	ISSUED FOR TENDER	2023/12/15	M.S.
1.	ISSUED FOR PERMIT	2023/11/20	M.S.

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FLOOR PLANS :
HVAC PIPING PLAN

SCALE : 3/16" = 1'-0"

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE
LOCATION : KOC DORM BLDG.

DRAWN BY	M.S.B.	PROJECT NO.	22-174
CHECKED BY	M.N.	DWG. NO.	M2.2
DATE	AUGUST 2023		
ISSUED	AUGUST 2023		

[illegible]

ERV SCHEDULE												
TAG	SERVICE	MAKE	CFM	MODEL	IN/OUT DUCT SIZE (IN)	ELECTRICAL DATA				WEIGHT (LB)	WxLxH (in)	REMARKS
						UNIT FLA (AMP.)	MCA (AMP.)	MOCp (AMP.)	ELECTRICAL V/PH/Hz			
ERV-1	BEDROOM'S	NU-AIRE	400	NU500-C	6x4 / 6x4	4.4	5.5	15	115 / 1 / 60	170	24 x 36 x 16	
C/W PROVIDE SPEED CONTROLLER, PUSH BUTTON OVERRIDE												

FAN SCHEDULE (EXISTING)											
TAG	SERVICE AND LABEL	MAKE	MODEL	VOLUME FLOW (CFM)	HP/WATT	E.S.P. (IN.)	MOTOR (RPM)	V	PH	HZ	REMARKS
EF-1	WASHROOM - MALE/FEMALE	PENN BARRY	-	-							EXISTING CEILING MOUNTED FAN
EF-2	WASHROOM - MALE/FEMALE	PENN BARRY	-	-							EXISTING CEILING MOUNTED FAN
EF-3	WASHROOM - MALE/FEMALE	PENN BARRY	-	-							EXISTING CEILING MOUNTED FAN

SPLIT HEAT PUMP SCHEDULE (HORIZONTAL)																		
TAG NO.	SERVICE AREAS	MAKE MODEL	COOLING CAPACITY (TR)	COOLING CAPACITY (KW)	HOT WATER HEATING CAPACITY KW	HOT WATER HEATING CAPACITY MBH	AIR FLOW (CFM)/(L/S)	ESP (INCH)/(Pa)	HOT WATER FLOW RATE GPM/LPS	WATER P. DROP (FT)/(M)	ELECTRICAL				DX COIL	CONDENSING UNIT	WEIGHT (Kg)	REMARKS
											FLA	VOLTS	MCA/HP	FUSE SIZE				
FCU 1	CAFE 02 & LOBBY 05	TRANE GAMSBOA48	2.5 Tons	102.4	163.82 (@ 60°C E.W.T.)	48.0 MBH (@ 140°F E.W.T.)	1100/518	0.35/87.5	6.5/0.41	9.26/2.82	6.0	208/1/60	3/4HP	15	CC-C	CU-1	85	SPLIT HEAT PUMP C/W BACKUP HOT WATER HEATING COIL
FCU 2	BEDROOM 7,8,9,14,15	TRANE GAMSBOA48	3.5 Tons	102.4	183.82 (@ 60°C E.W.T.)	48.0 MBH (@ 140°F E.W.T.)	1100/518	0.35/87.5	6.5/0.41	9.26/2.82	8.0	208/1/60	3/4HP	15	CC-C	CU-3	85	SPLIT HEAT PUMP C/W BACKUP HOT WATER HEATING COIL

NOTE:

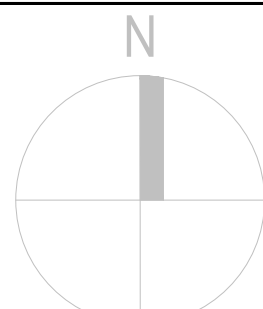
- COOLING CAPACITY IS BASED ON 86 DEG F E.W.T, 80/67 AIR ON COIL.
- HEATING CAPACITY IS BASED ON 64 DEG F E.W.T., 70 DEG F AIR ON COIL.
- FOR EACH FCU, RUN CONDENSATE TO THE NEAREST DRAIN.

HEAT PUMP OUTDOOR UNIT SCHEDULE										
TAG	SERVED	TONNAGE	TC CAPACITY (KW)	SUPPLY POWER			MAKE	MODEL	WEIGHT (Kg)	REMARKS
				V/PH/HZ	MCA	MOCP				
CU-1	—	2.5	109.2	208/1/60	16	25	TRANE	A4AC3030A	74	
CU-2	—	3.5	109.2	208/1/60	21	30	TRANE	A4AC3030A	74	

CIRCULATING PUMP SCHEDULE										
TAG	SYSTEM SERVED	SIZE	FLOW (GPM)	HEAD (FT)	MOTOR			DUTY	MAKE & MODEL	REMARKS
					HP	R.P.S.	V/PH/HZ			
P-1	BOILER PUMP	3X3X10	20.0	30.0	7.46	30	208/1/60	RUNNING	ARMSTRONG 4300	
P-2	BOILER PUMP	3X3X10	20.0	30.0	7.46	30	208/1/60	STAND BY	ARMSTRONG 4300	

HVAC SYMBOL LEGEND	
	THERMOSTAT
	SQUARE DIFFUSER
	12"x12" / 300x300 RETURN AIR GRILLE
	24"x12" / 600x300 RETURN AIR GRILLE
	24"x24" / 600x600 RETURN AIR GRILLE
	SUPPLY AIR DUCT UP/DN
	RETURN/EXHAUST AIR DUCT UP/DN
	RECTANGULAR DUCT W/ TAKE-OFF & BALANCING DAMPER
	CIRCULAR DUCT W/ TAKE-OFF & BALANCING DAMPER
	FLEX DUCT W/ TAKE-OFF & BALANCING DAMPER
	CIRCULAR DUCT (DIAMETER)
	RECTANGULAR DUCT (WIDTH x HEIGHT)
	OPEN ENDED DUCT
	ONE WIDTH RADIUS DUCT ELBOW
	MITERED ELBOW DUCT W/ TURNING VANES
	RECTANGULAR DUCT CONCENTRIC TRANSITION
	RECTANGULAR DUCT ECCENTRIC TRANSITION
	RECTANGULAR DUCT ELEVATION TRANSITION
	CIRCULAR DUCT CONCENTRIC TRANSITION
	CIRCULAR DUCT ECCENTRIC TRANSITION
	CIRCULAR TO RECTANGULAR DUCT TRANSITION
	DUCTS ABOVE CROSSING DUCTS
	DUCTS BELOW CROSSING OTHER DUCTS
	ACOUSTICALLY LINED TRANSFER AIR DUCT W/ GRILLE
	ACOUSTICALLY LINED TRANSFER AIR DUCT
	SIDE WALL SUPPLY AIR GRILLE
	SIDE WALL RETURN AIR GRILLE
	CEILING MOUNTED RETURN AIR GRILLE
	CEILING MOUNTED SUPPLY AIR GRILLE
	# = NUMBER OF TYPICAL GRILLES L = LENGTH OF GRILLE W = WIDTH OF GRILLE FLOW = AIR FLOW AMOUNT
	# = NUMBER OF TYPICAL DIFFUSER ø = NECK CONNECTION SIZE/FLEX DUCT SIZE T = TYPE OF DIFFUSER FLOW = AIR FLOW AMOUNT
	EQ # = EQUIPMENT # FLOW = AIR FLOW OUT/IN
	WALL MOUNTED FORCE FLOW HEATER
	CEILING MOUNTED FORCE FLOW HEATER
	VERTICAL/PACKAGED TERMINAL AIR CONDITIONING UNIT UNIT #

1. UNLESS OTHERWISE NOTED ON DRAWINGS, ALL EXHAUST FANS AND ELECTRIC HEATERS IN EXISTING WASH ROOMS TO REMAIN.
2. FOR WORK RELATED TO STRUCTURAL, ELECTRICAL, AND ARCHITECTURAL REFER TO RESPECTIVE DRAWINGS FOR DETAILS.
3. INSTALL NEW SPLIT HEAT PUMPS, PTACK UNITS, ERV, FAN & SUPPLY AIR GRILLES AT THE POSITIONS AS SHOWN ON DRAWING.
4. MAINTAIN ALL FIRE SEPARATIONS. PROVIDE DAMPERS AS REQUIRED.
5. CONTRACTOR'S RESPONSIBLE FOR REVIEWING SITE AND NOTING CONDITION & LOCATION OF THE EXISTING FACILITIES RELATED TO THE SCOPE PRIOR TO STARTING WORK.
6. BALANCE FLOW TO EACH BRANCH & DIFFUSER. ADD BALANCING DAMPER AS REQUIRED.



22-174

PROJECT NORTH

5.	DRAWING UPDATED		2026/10/07	M.S.
4.	UPDATED AS PER LATEST ARCH.		2025/09/09	M.S.
3.	UPDATED AS PER LATEST ARCH.		2025/09/09	M.S.
2.	ISSUED FOR TENDER		2023/12/15	M.S.
1.	ISSUED FOR PERMIT		2023/11/20	M.S.
NO.	REVISIONS		DATE	BY

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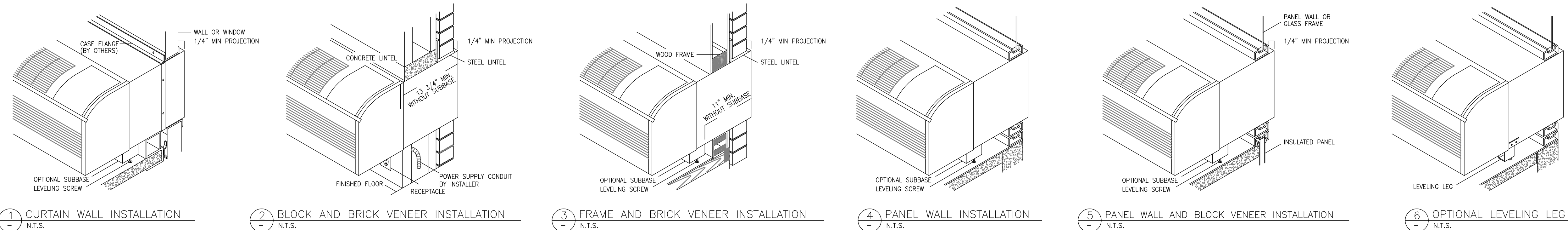
This drawing must be read in the context of all the other drawings which constitute the document.

SCALE : 3/16" = 1'-0"

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE

LOCATION : KOC DORM BLDG.

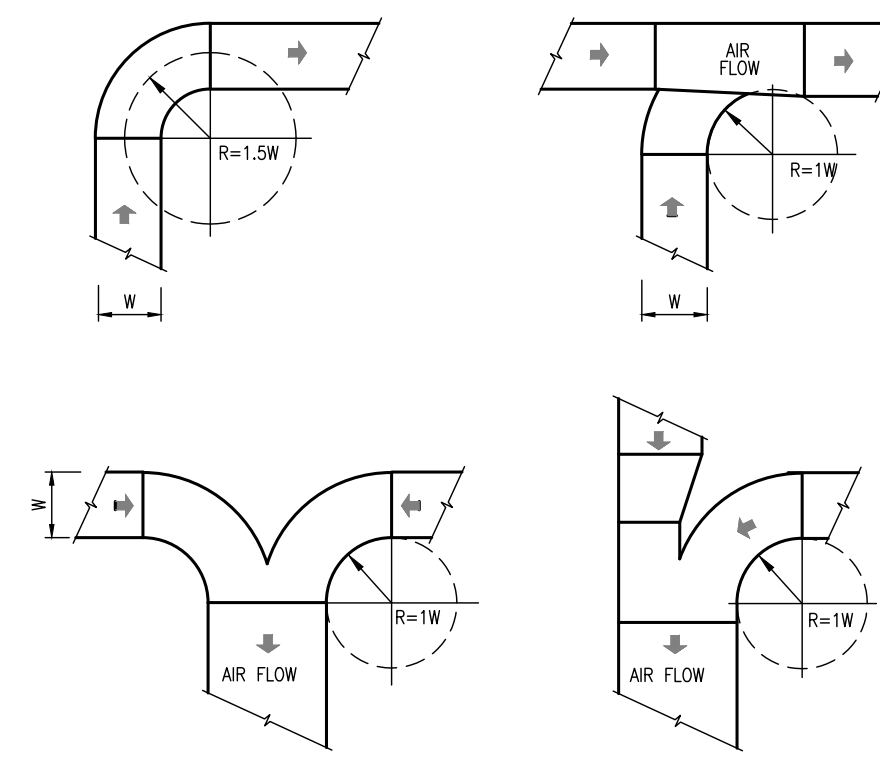
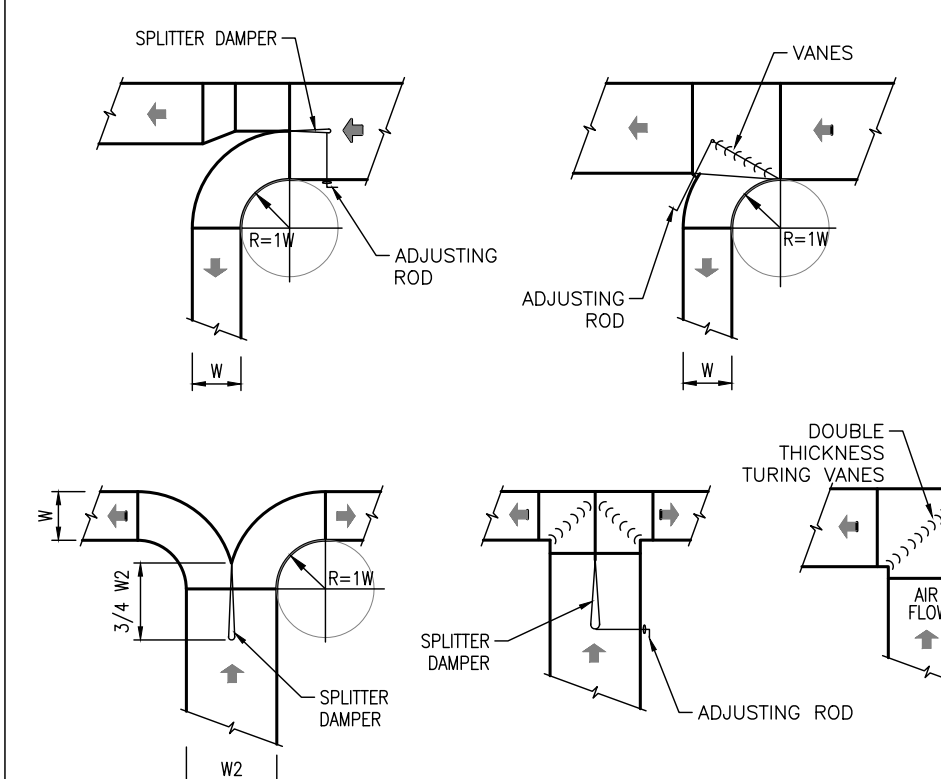
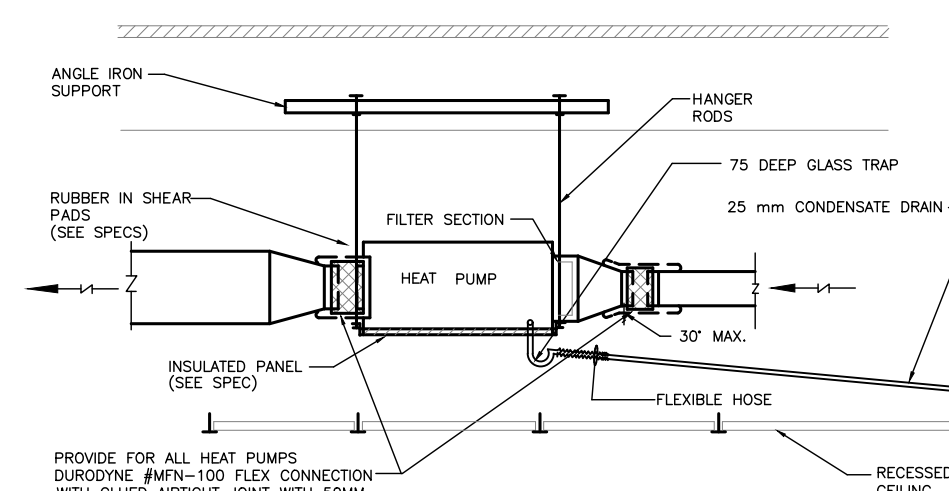
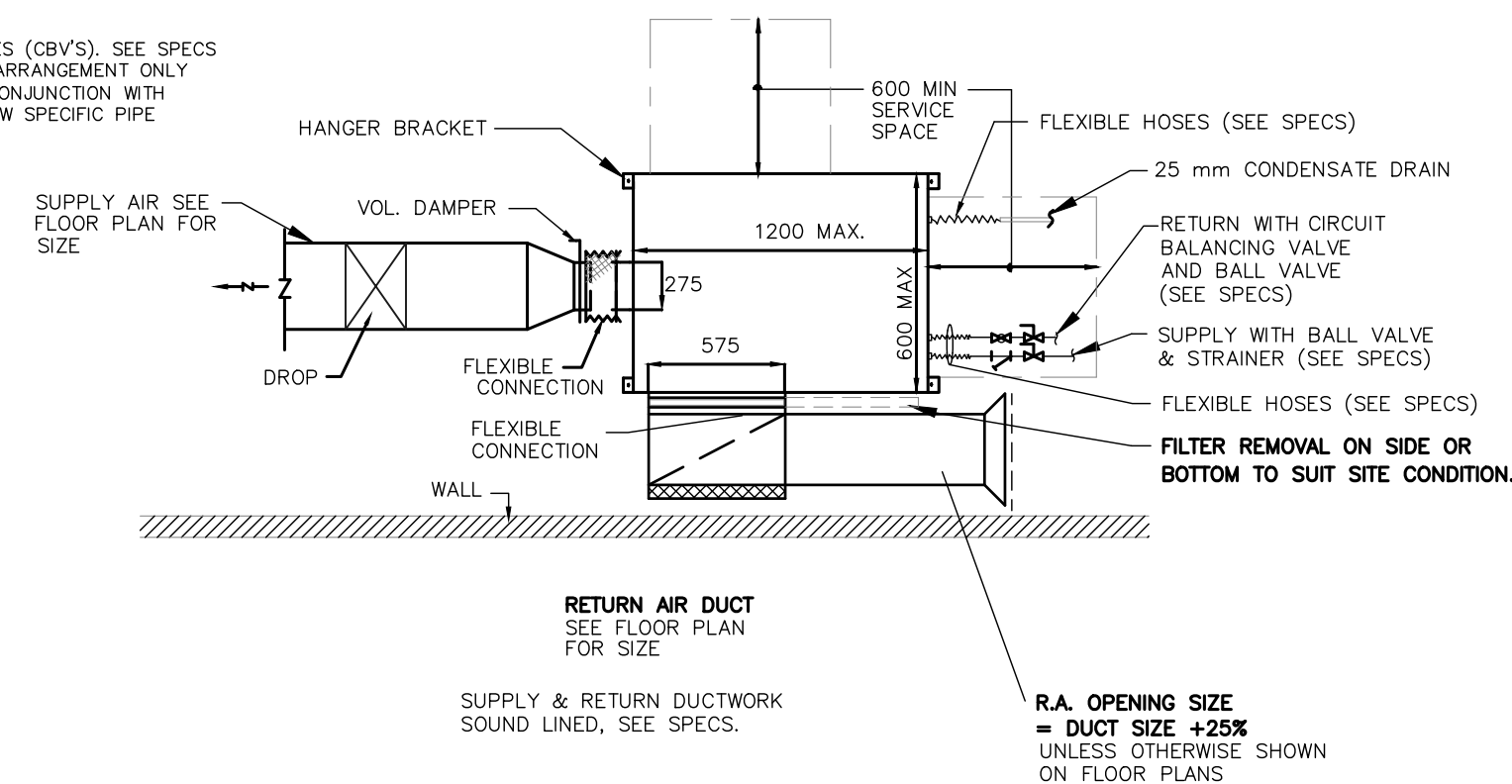
DRAWN BY	M.S.B.	PROJECT NO. 22-174 DWG. NO. M2.3
CHECKED BY	M.N.	
DATE	AUGUST 2023	
ISSUED	AUGUST 2023	



PTAC INSTALLATION DETAIL
N.T.S.

NOTES

1. CIRCUIT BALANCING VALVES (CBV'S). SEE SPEC.
2. DETAILS SHOW GENERAL ARRANGEMENT ONLY AND MUST BE READ IN CONJUNCTION WITH FLOOR PLANS WHICH SHOW SPECIFIC PIPE CONNECTIONS ETC.

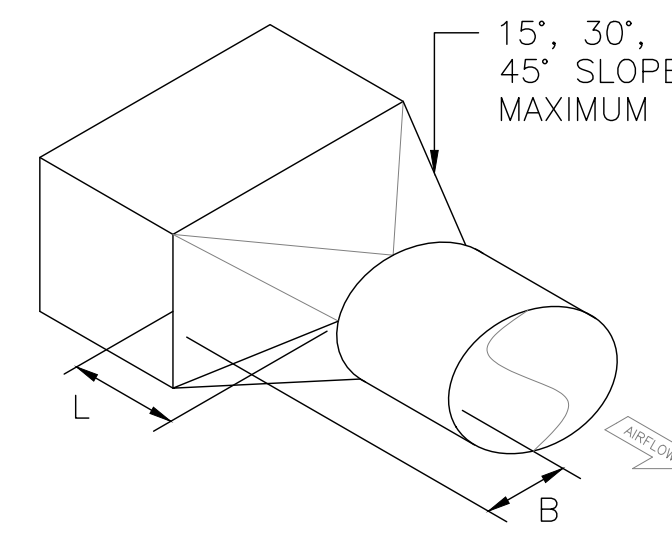
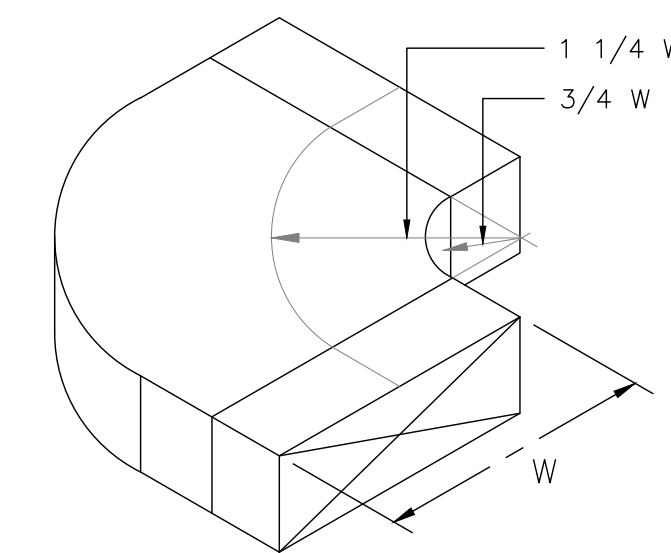
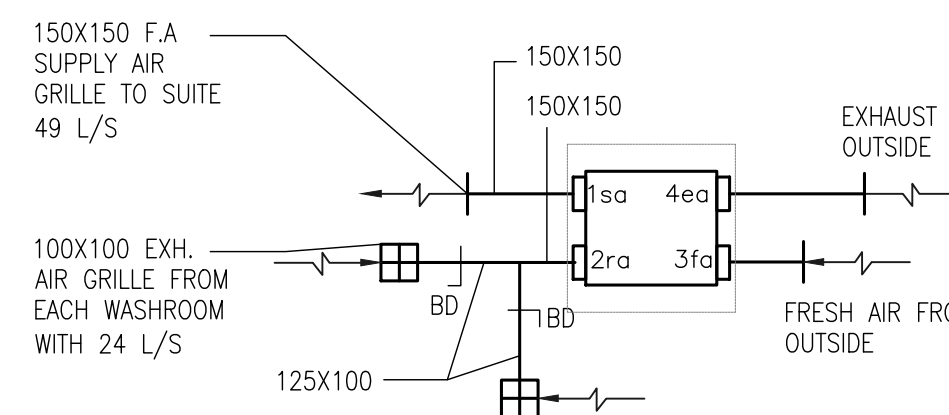
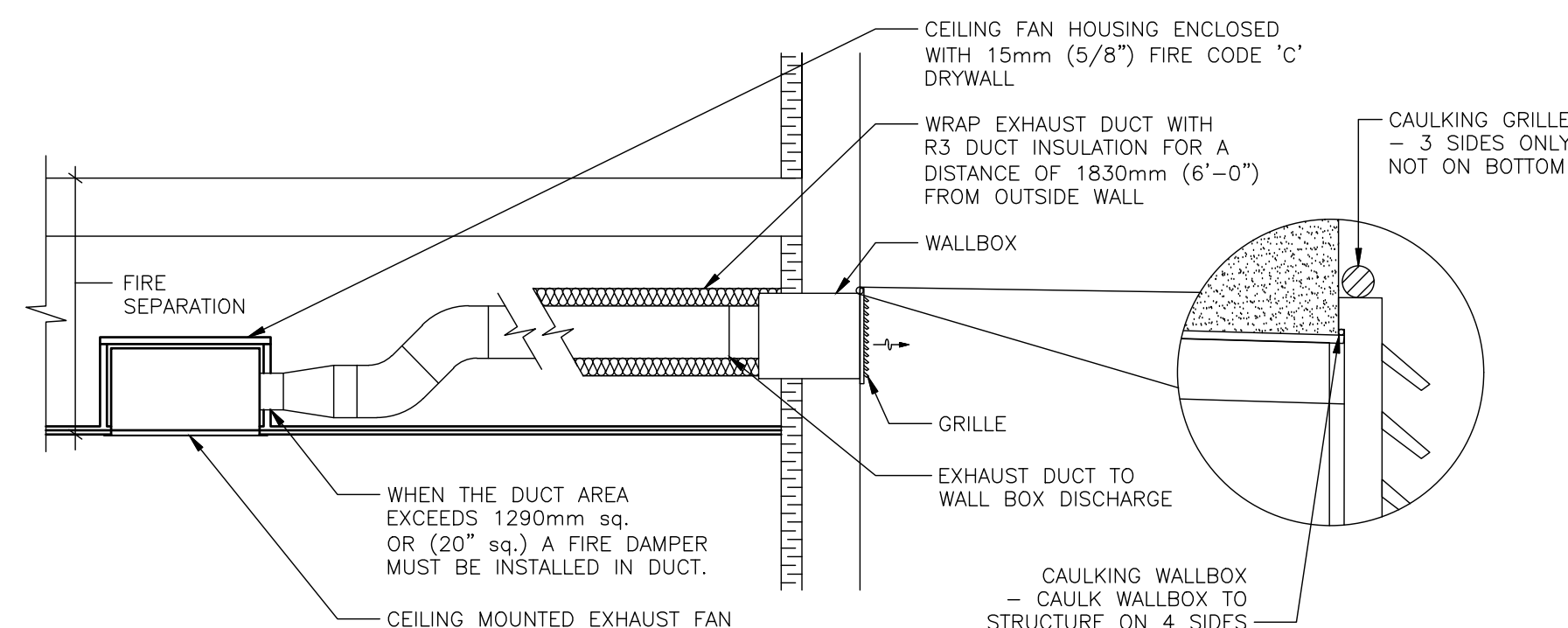


TYPICAL HORIZONTAL HEAT PUMP DETAIL
N.T.S.

SECTION THRU TYPICAL HORIZONTAL HEAT PUMP
N.T.S.

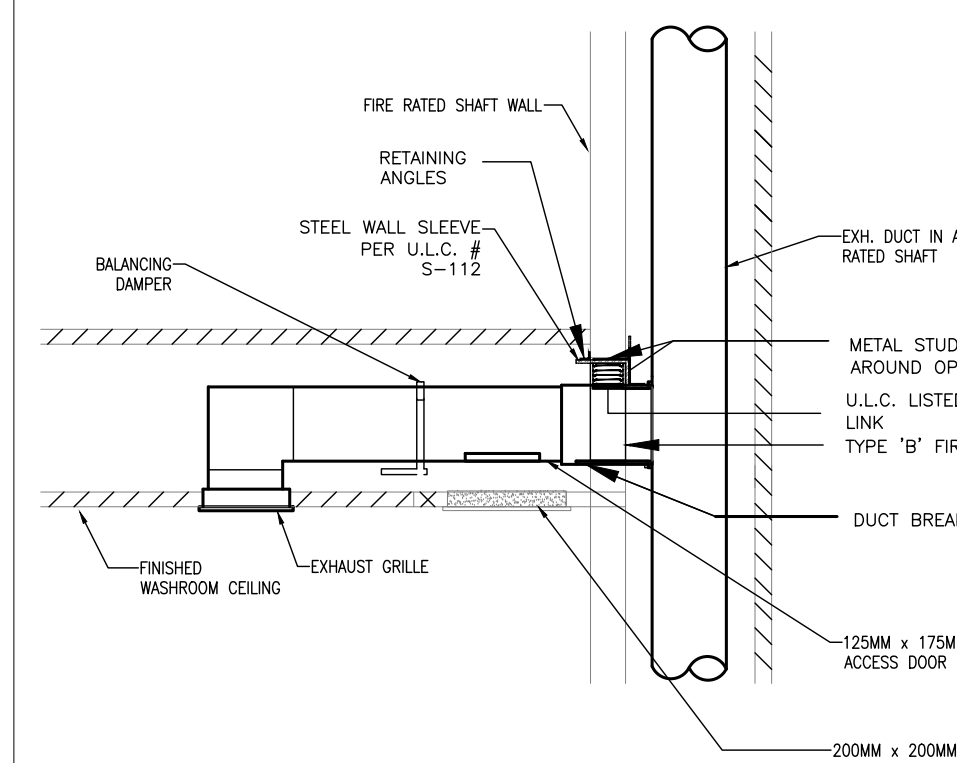
TYPICAL SUPPLY DUCT BRANCH
N.T.S.

TYPICAL EXHAUST & RETURN DUCT BRANCH
N.T.S.

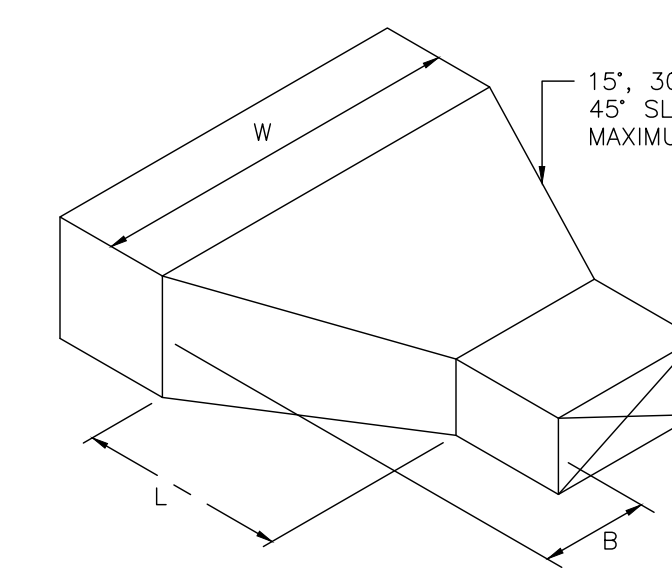
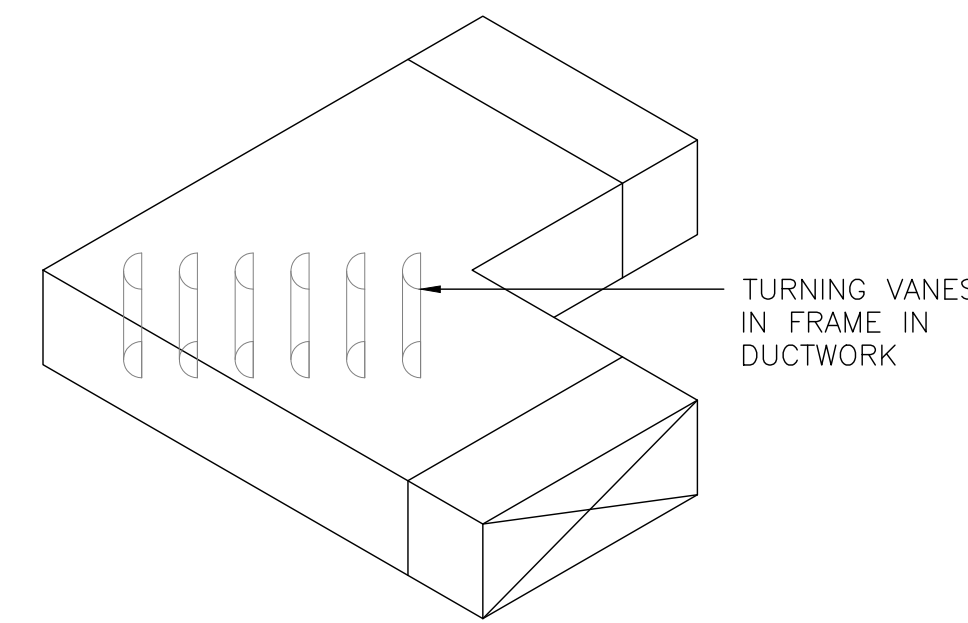
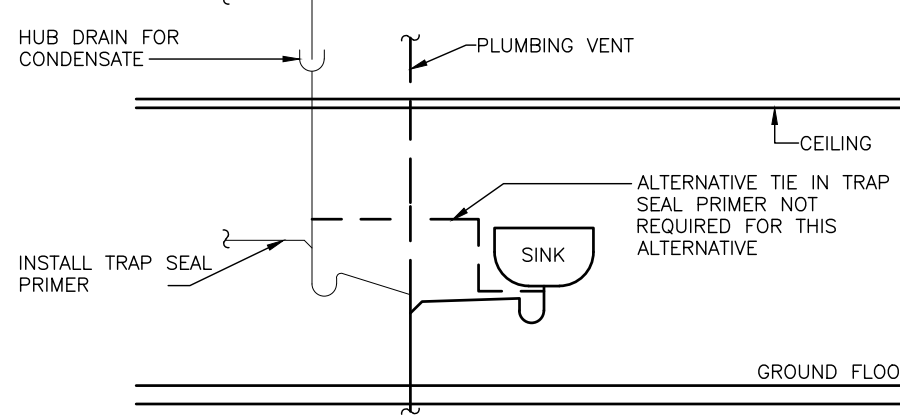
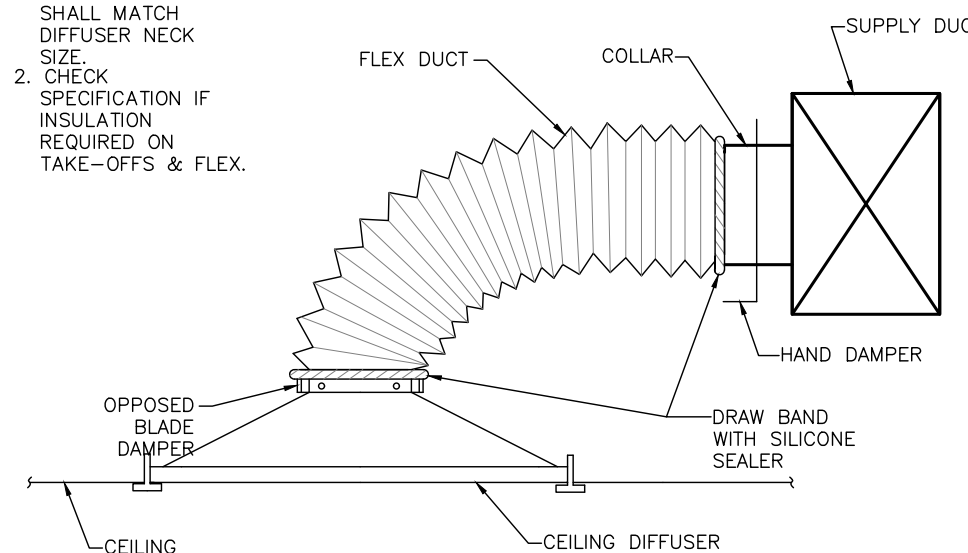


TYPICAL CEILING EXHAUST FAN DETAIL FOR FIRE RATED INSTALLATION
N.T.S.

TYPICAL ERV AIR DISTRIBUTION
N.T.S.



- NOTE:
1. MAX. FLEX DUCT LENGTH IS 1800mm
FLEX DUCT SIZE SHALL MATCH DIFFUSER NECK SIZE.
 2. CHECK SPECIFICATION IF INSULATION REQUIRED ON TAKE-OFFS & FLEX



NOTE:
PROVIDE TURNING VANES WHERE
CENTERLINE RADIUS IS $< 1 \frac{1}{4}W$

TYPICAL WASHROOM EXHAUST INSTALLATION
N.T.S.

TYPICAL DIFFUSER CONNECTION
N.T.S.

DETAIL OF INDIRECT DRAIN FOR CONDENSATE
NTS.

DUCT ELBOWS AND TRANSITIONS
N.T.S.

5.	DRAWING UPDATED	2026/10/07	M.S.
4.	UPDATED AS PER LATEST ARCH.	2025/09/09	M.S.
3.	UPDATED AS PER LATEST ARCH.	2025/09/03	M.S.
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FLOOR PLANS :
STANDARD DETAILS

SCALE : $3/16" = 1'-0"$

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE

LOCATION : KOC DORM BLDG

DRAWN BY	M.S.B.	PROJECT NO.	22-174
CHECKED BY	M.N.		
DATE	AUGUST 2023	DWG. NO.	M2.4
ISSUED	AUGUST 2023		

ELECTRICAL SPECIFICATIONS

1. GENERAL CONDITIONS

- A. THE ELECTRICAL CONTRACTOR SHALL FURNISH ALL LABOUR, MATERIAL, TOOLS, EQUIPMENT, ETC. REQUIRED TO COMPLETE ALL WORK SHOWN ON THE DRAWINGS AND HEREIN SPECIFIED. THE WORK SHALL BE IN ACCORDANCE WITH RULES AND REGULATIONS OF ALL AUTHORITIES HAVING LEGAL JURISDICTION OVER THE WORK. THIS CONTRACTOR SHALL PROVIDE ANY SMALL ITEMS OF WORK NOT SPECIFICALLY CALLED FOR BUT REQUIRED TO COMPLETE THE INTENDED INSTALLATION.
- B. THE ENGINEER RESERVES THE RIGHT TO APPROVE THE QUALITY OF MATERIAL AND WORKMANSHIP, ALSO TO CALL FOR ANY TESTS WHICH ARE DEEMED NECESSARY DURING THE PROGRESS OF THE WORK AND A COMPLETE TEST OF EACH SYSTEM AT THE COMPLETION OF THE WORK. THE COST OF SUCH TESTS ARE NOT TO BE CONSIDERED AS EXTRAS.

2. SCOPE OF WORK THE WORK SHALL CONSIST OF, BUT SHALL NOT BE LIMITED TO THE FOLLOWING:

- A. ALL NECESSARY CONDUIT WIRING AND CONNECTIONS FOR A COMPLETE INSTALLATION.
- B. LIGHTING AND POWER DISTRIBUTION SYSTEM AS SHOWN ON THE DRAWINGS.
- C. MOTOR CIRCUITS.
- D. COMPLETE CONDUIT SYSTEM FOR LOW VOLTAGE SYSTEM WHERE NOTED ON THE DRAWINGS.

3. CODES, PERMITS AND INSPECTION

- A. THIS CONTRACTOR SHALL BE RESPONSIBLE FOR AND SHALL OBTAIN ALL PERMITS, INSPECTIONS, ETC. AS REQUIRED BY ALL AUTHORITIES HAVING JURISDICTION OVER THIS WORK AND SHALL PAY FOR SAME. THESE COSTS SHALL BE INCLUDED IN THE TENDER PRICE.
- B. ALL PERMITS SHALL BE DELIVERED TO THE OWNER REPRESENTATIVE AS SOON AS THEY BECOME AVAILABLE.

4. DRAWINGS

5. THIS CONTRACTOR SHALL PREPARE, AT HIS OWN EXPENSE, ANY LARGE SCALE WORKING DRAWINGS WHICH MAY BE REQUIRED BY THE EXAMINING AUTHORITIES.

6. AS-BUILT DRAWINGS

7. THIS CONTRACTOR SHALL KEEP A SEPARATE SET OF WHITE PRINTS ON THE SITE AND NOTE ALL CHANGES AND DEVIATIONS FROM THE ORIGINAL DESIGN. TWO SETS OF THESE PLANS ALONG WITH UPDATED ACAD 2000 DRAWINGS SHOWING ALL AS-BUILT CONDITIONS SHALL BE FORWARDED TO THE ENGINEER AT THE COMPLETION OF THIS CONTRACT BEFORE APPLYING FOR FINAL PAYMENT. SUBMIT TO ENGINEERS THE FOLLOWING DOCUMENTS FOR REVIEW AND APPROVAL AT THE COMPLETION OF PROJECT: – ESA INSPECTION CERTIFICATE.

8. SHOP DRAWINGS SUBMIT SIX (6) COPIES OF MANUFACTURER’S SHOP DRAWINGS FOR ALL DEVICES, SYSTEMS AND SUCH FOR REVIEW BY THE ENGINEER.

9. EXAMINATION OF SITE

- A. THIS CONTRACTOR SHALL VISIT THE SITE OF THE PROJECT AND FAMILIARIZE HIMSELF WITH THE SPECIFIC SITE CONDITIONS.

- B. ANY DEVIATION AND / OR CONFLICTS ON SITE ESPECIALLY CONCERNING THE DISTRIBUTION DIAGRAM (S) AS SHOWN ON THE PLANS, SHALL BE REPORTED TO THE ENGINEER PRIOR TO SUBMITTING TENDER.

8. CONSTRUCTION SCHEDULE THIS CONTRACTOR SHALL SCHEDULE AND PERFORM HIS WORK TO MEET THE COMPLETION SCHEDULE AS SET OUT BY THE PLANNING HOUSE.

9. REVISIONS AND EXTRAS NO ADDITIONAL MONEY OVER THE CONTRACT PRICE SHALL BE PAID UNLESS AN APPROVED CHANGE ORDER IS ISSUED BY THE ENGINEER. CLAIMS FOR EXTRAS SHALL BE SUBMITTED WITH A COMPLETE BREAKDOWN OF MATERIAL, LABOUR, HOURLY RATES, ETC. THERE SHALL BE NO EXTRA CLAIM FOR RELOCATION OF ANY EQUIPMENT WITHIN 10 FEET FROM ORIGINAL LOCATION.

10. CLEAN UP THIS CONTRACTOR SHALL BE RESPONSIBLE TO PERIODICALLY REMOVE ALL DEBRIS AND TO KEEP THIS AREA CLEAN AT ALL TIMES.

11. CUTTING AND PATCHING ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR ALL CUTTING AND PATCHING, OPENINGS, ETC. AND SHALL INFORM THE GENERAL CONTRACTOR OF SLEEVES OR OPENINGS REQUIRED DURING TENDERING SO THAT COSTS ARE INCLUDED AND SHALL INFORM THE GENERAL CONTRACTOR IN SUFFICIENT TIME TO PREVENT UNNECESSARY CUTTING.

12. CONDUIT IN SLAB THIS CONTRACTOR SHALL NOT INSTALL RIGID CONDUIT AND FLOOR BOXES IN STRUCTURE FLOOR SLABS OR FRAMING OR CUT STRUCTURAL SLABS AND FRAMING FOR THIS PURPOSE IN SUCH A MANNER THAT THE STRUCTURAL INTEGRITY OF THE FLOOR IS WEAKENED. COORDINATE SUCH WORK WITH THE ARCHITECT OR STRUCTURAL ENGINEER.

13. DELIVERY DATES THIS CONTRACTOR SHALL PLACE AN ORDER FOR ALL MATERIAL AND EQUIPMENT IMMEDIATELY AFTER SIGNING OF THE CONTRACT. HE SHALL SUBMIT A LIST OF DELIVERY DATES FOR EACH TYPE OF EQUIPMENT WITHIN 30 DAYS OF THE AWARDED OF THE CONTRACT. THE LIST SHALL INCLUDE MANUFACTURER’S NAMES.

14. EQUIPMENT AND MATERIAL GROUNDING SHALL BE AS REQUIRED BY ELECTRICAL CODE AND THE APPROVAL OF ALL EQUIPMENT AND MATERIAL UNLESS SPECIFICALLY NOTED OTHERWISE, SHALL BE NEW AND WITHOUT BLEMISH OR DEFECT. ALL MATERIAL AND EQUIPMENT SHALL BE OF THE TYPE SUBJECT TO FACTORY MUTUAL, "UNDERWRITERS LABORATORIES OF CANADA" OR "CANADIAN STANDARDS ASSOCIATION INSPECTION AND APPROVAL" AND SHALL BEAR "U.L.C." OR "C.S.A." LABELS.

16. IDENTIFICATION PROVIDE LAMACOID IDENTIFICATION NAMEPLATES. THESE SHALL BE BLACK WITH WHITE ENGRAVED LETTERS AND SHALL BE INSTALLED WITH SCREWS ON ALL EQUIPMENT, DISCONNECT SWITCHES, PANELS, ETC. INDICATING THE LOAD SERVED. EACH LIGHTING PANEL SHALL HAVE A TYPEWRITTEN DIRECTORY SHOWING LIGHTS OR EQUIPMENT CONNECTED TO EACH CIRCUIT. DIRECTORIES SHALL BE MOUNTED ON THE INSIDE OF THE PANEL DOOR WITH A TRANSPARENT PLASTIC COVER. RECEPTACLES AND LIGHT SWITCHES SHALL BE LABELLED WITH CIRCUIT NUMBER BY STICK-ON APPROVED TYPE LABELLING.

17. COORDINATION

THIS CONTRACTOR SHALL BE RESPONSIBLE TO CO-ORDINATE THE INSTALLATION OF EQUIPMENT, CONDUIT WORK, LIGHTING FIXTURES, ETC. WITH OTHER TRADES PRIOR TO THE ACTUAL INSTALLATION.

18. ACCESSIBILITY

ALL WORK SHALL BE INSTALLED SO AS TO BE READILY ACCESSIBLE FOR OPERATION, MAINTENANCE AND REPAIRS.

19. RESPONSIBILITY THIS TRADE SHALL BE RESPONSIBLE FOR THIS WORK UNTIL THE COMPLETION AND FINAL ACCEPTANCE, FOR REPLACING ANY ITEM THAT MAY BE DEFECTIVE, DAMAGED, LOST OR STOLEN WITHOUT ADDITIONAL COST TO THE OWNER OR DELAY TO THE COMPLETION OF THE PROJECT.

20. WARRANTEE

THIS CONTRACTOR SHALL WARRANT ALL WORK AND EQUIPMENT INSTALLED UNDER THIS CONTRACT, AGAINST ALL DEFECTS OR WORKMANSHIP AND MATERIAL FOR A PERIOD OF ONE YEAR AFTER ACCEPTANCE OF THE INSTALLATION BY THE OWNER.

21. CONDUIT AND FITTINGS

- A. CONDUIT SIZES SHALL BE AS INDICATED ON THE DRAWINGS AND SHALL NOT BE REDUCED IN SIZE WITHOUT AUTHORIZATION. CONDUIT IN FINISHED AREA SHALL BE CONCEALED. ALL CONDUIT SHALL BE INSTALLED PARALLEL TO BUILDING LINES. MAKE FINAL CONNECTIONS TO VIBRATING EQUIPMENT WITH FLEXIBLE CONDUIT.
- B. CONDUITS SHALL BE INSTALLED AT A MINIMUM OF 6" (152MM) FROM UNINSULATED HEATING PIPES.
- C. CLEAN INTERIOR OF ALL CONDUITS TO REMOVE WATER AND DEBRIS BEFORE PULLING WIRES.
- D. BX CABLES SHALL BE USED FOR FINAL SHORT CONNECTIONS BETWEEN OUTLET AND OUTLET BOX IN CEILING SPACE TO CEILING MOUNTED LIGHTING FIXTURES OR TO FEED OUTLETS; RECEPTACLES IN THE DRYWALL PARTITIONS. BX CABLES ARE NOT ALLOWED TO RUN IN THE OPEN CEILING AREAS.
- E. EMPTY FLEXIBLE STEEL CONDUIT 1/2" (13MM) MINIMUM DIAMETER AND A MAXIMUM OF 6 FEET (1830MM) OR (1.83M) LONG, WITH A FISH WIRE AS INDICATED IN THE DRAWINGS CAN ALSO BE USED.
- F. FOR FIREPROOFING WHERE HOLES ARE DRILLED THROUGH FLOORS OR WALLS, THEY SHALL BE PACKED AROUND CONDUIT WITH "3M" BRAND FIRE BARRIER CAULK FB369 COMPOUND. MINIMUM CONDUIT SHALL BE 3/4" (19MM), UNLESS OTHERWISE NOTED.
- G. CONDUITS AND ARMORED CABLES SHALL BE SUPPORTED INDEPENDENTLY OF THE SUSPENDED CEILING.
- H. SUPPLY AND INSTALL CONDUIT FOR WIRING AND CABLES WHERE THERE IS NO SUSPENDED CEILING SPACE.

22. PULL AND JUNCTION BOXES

- A. BOXES SHALL BE CODE GAUGE AND SIZES TO MEET THE ELECTRICAL CODE REQUIREMENTS. SHEET STEEL BOXES FOR CONCEALED WORK AND CAST BOXES FOR EXPOSED WORK.
- B. PROVIDE BARRIERS IN BOXES WHERE DIFFERENT VOLTAGES ARE USED.
- C. PROVIDE PULL BOXES ON CONDUIT AT 50-FOOT (15.25M) INTERVALS.
- D. BOXES SHALL BE SUPPORTED INDEPENDENTLY OF CONDUIT.

23. INSTALLATION OF OUTLETS

THE PLANS SHOW APPROXIMATE LOCATION OF OUTLETS, EXACT LOCATION SHALL BE CO-ORDINATED ON THE SITE WITH OTHER TRADES, ARCHITECTURAL PLANS, ETC. OUTLETS INACCURATELY LOCATED SHALL BE RE-ADJUSTED OR

RELOCATED AT THE CONTRACTOR’S EXPENSE. UNLESS OTHERWISE NOTED ON THE PLAN (S), LOCATE OUTLETS AS FOLLOWS:

- A. RECEPTACLES, TELEPHONE OUTLETS 18" ABOVE FINISHED FLOOR.
- B. OUTLETS OVER COUNTER, 42" ABOVE FLOOR OR CO-ORDINATE ON JOB.
- C. OUTLETS IN MECHANICAL, ELECTRICAL AND TELEPHONE ROOMS 47" (1200MM) ABOVE FLOOR.

24. WIRE AND CABLE

- A. UNLESS OTHERWISE NOTED, ALL WIRES SHALL BE COPPER R90, RW90, RA90 OR RWU90 RATING AS REQUIRED ON SITE AND AS DICTATED BY CODE. ALUMINUM WIRE SHALL NOT BE USED.
- B. THE MINIMUM PERMISSIBLE SIZE FOR BRANCH CIRCUIT WIRING SHALL BE #12 (20m). EXCEPT FOR EMERGENCY LIGHTING SYSTEM SHALL BE MIN. #10.
- C. FOR BRANCH WIRING EXCEEDING 100 FEET (30.5M) TO FURTHEST OUTLET FROM A PANEL SHALL BE #10 AT 120 VOLTS.
- D. WIRES AND CABLES SHALL BE RATED AT 600 VOLTS EXCEPT FOR LOW VOLTAGE CONTROL WIRING WHICH SHALL BE RATED AT 300V.
- E. ALL WIRES SHALL BE NEW AND DELIVERED TO THE SITE OF THE PROJECT IN THEIR ORIGINAL PACKING. WIRES #10 AND BIGGER SHALL BE STRANDED. THIS DENOTES GAUGE #8, #6, #4, ETC. WIRES SHALL BE FACTORY IDENTIFIED, SHOWING SIZE, VOLTAGE RATING AND INSULATION TYPE.
- F. PROVIDE SEPARATE INSULATED GROUND CONDUCTOR FOR EACH FEEDER AND BRANCH CIRCUIT.
- G. FINAL CONNECTIONS TO LUMINAIRES SHALL ORIGINATE FROM AN OUTLET BOX. CONNECTIONS OF FIXTURE BODY TO FIXTURE BODY SHALL NOT BE ACCEPTABLE.
- H. CODE APPROVED WIRE SHALL BE USED FOR FINAL LUMINAIRE OR APPLIANCE CONNECTIONS. 26.

WIRING DEVICES

- A. DUPLEX RECEPTACLES SHALL BE GROUNDING TYPE MINIMUM RATED FOR 15A. 120V, SPECIFICATION GRADE AND TOGGLE TYPE UNLESS OTHERWISE SPECIFIED ON PLANS. THESE SHALL HAVE BREAK-OFF LINE TO ALLOW FOR SPLIT WIRING OR 2 CIRCUITS. EQUIVALENT MANUFACTURER OF RECEPTACLES ARE AS FOLLOWS: "ARROW HART" AND "PASS & SEYMOUR", SPECIFICATION GRADE. GENERAL PURPOSE RECEPTACLE SHALL BE WHITE AND RECEPTACLES DEDICATED FOR ELECTRONIC EQUIPMENT SUCH AS COMPUTERS, COPIER ETC. SHALL BE WHITE IN COLOR.
- I. LIGHT SWITCHES SINGLE POLE/3-WAY AND 4-WAY AS SHOWN SHALL BE "WHITE" AND SPECIFICATION GRADE EQUAL TO HUBBELL 1200 SERIES. EQUIVALENT MANUFACURER OF SWITCHES ARE AS FOLLOWS: "ARROW HART" AND "PASS & SEYMOUR", SPECIFICATION GRADE.

TESTING, VERIFICATION & INTEGRATION OF LIFE SAFETY SYSTEMS

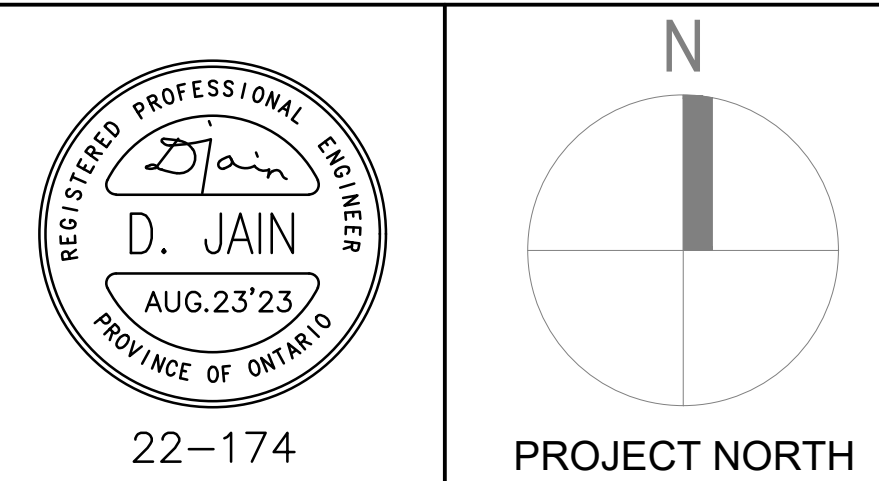
- A. ALL LIFE SAFETY SYSTEMS INTEGRATED WITH THE FIRE ALARM SYSTEM MUST BE VERIFIED AND DOCUMENTED IN ACCORDANCE WITH O.B.C. 3.2.10.1 AND THE LATEST EDITION OF CAN/ULC-S1001-11, "STANDARD FOR INTEGRATED SYSTEMS TESTING OF FIRE PROTECTION AND LIFE SAFETY SYSTEMS" THE FOLLOWING IS TO BE PREPARED AND SUBMITTED BY THE ELECTRICAL CONTRACTOR AND SIGNED BY A P.ENG. TO THE CONSULTANT FOR REVIEW AT THE START OF THE PROJECT:
- B. INTEGRATED TESTING PLAN – A WRITTEN SPECIFIC DOCUMENT PREPARED BY THE INTEGRATED TESTING COORDINATOR, OUTLINING THE REQUIRED TESTS AND NECESSARY FUNCTIONAL RESULTS TO CONDUCT INTEGRATED FIRE PROTECTION AND LIFE SAFETY SYSTEMS TESTING.
- C. INTEGRATED TESTING REPORT – A WRITTEN PROJECT SPECIFIC DOCUMENT, PREPARED BY THE INTEGRATED TESTING COORDINATOR, DOCUMENTING THE IMPLEMENTATION OF THE INTEGRATED TESTING PLAN.

THE FOLLOWING SYSTEMS (WHERE APPLICABLE) ARE TO BE INTEGRATED INTO THE TESTING PLAN AND TESTING REPORT:

SEQUENCE OF OPERATION)	.1	FIRE ALARM SYSTEM (INCLUDING
	.2	MASS NOTIFICATION SYSTEM
	.3	ELEVATORS
	.4	EMERGENCY GENERATORS
	.5	AUDIO/VISUAL SYSTEM
	.6	LIGHTING CONTROL SYSTEM
	.7	NOTIFICATION SYSTEMS
	.8	FIRE PROTECTION SYSTEM INCLUDING
SPRINKLER SYSTEM, STANDPIPE, ETC.	.9	FREEZE PROTECTION SYSTEMS
	.10	FIRE SUPPRESSION SYSTEMS
	.11	SMOKE CONTROL PRESSURIZATION
SYSTEMS	.12	SMOKE CONTROL SMOKE EXHAUST
SYSTEMS	.13	HAZARDOUS PROTECTION MONITORING
	.14	SMOKE ALARMS
	.15	NURSE CALL SYSTEM
	.16	OTHER SYSTEM (WHERE APPLICABLE)

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3	ADDENDUM EA-2	2026/10/07	DJ
3	ISSUED FOR PERMIT/TENDER	2025/09/25	DJ
2	ISSUED FOR COORDINATION	2025/08/18	DJ
1	ISSUED FOR PERMIT	2023/09/15	DK
NO.	REVISIONS	DATE	BY

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FLOOR PLANS :

SPECIFICATIONS

SCALE : 3/16" = 1'-0"

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE

LOCATION : KOC DORM BLDG.

DRAWN BY	DK	PROJECT NO.	22-174
CHECKED BY	DJ	DWG NO.	E001
DATE	AUGUST 2025		
ISSUED	AUGUST 2025		

MECHANICAL EQUIPMENT STARTER AND ELECTRICAL DATA SCHEDULE

EQUIPMENT & LABEL		ELECTRICAL DATA				STARTER TYPE	THERMOSTAT	ON/OFF SEL. SWITCH IN COVER	START/STOP P.B. IN COVER	H/O/A SELECTOR SWITCH IN COVER	120V. CONTROL TRANSFORMER	REMOTE P.B. & P.L. STATION	REMOTE SEL. S. & P.L. STATION	INTERLOCK WITH	WIRING BY MECHANICAL	REMARKS
		LOAD (KW/MCA/HP)	PHASE	VOLTAGE	MOC											
FCU	FCU (TYP OF 2)	MCA 6,8	1	240	15	INTEGRAL	●			●	●				●	ELECTRICAL DIVISION WILL WIRE TO UNIT DISCONNECT
CU	CU (TYP OF 2)	SEE MECH	1	240	25,30	INTEGRAL	●			●	●				●	ELECTRICAL DIVISION WILL WIRE TO UNIT WEATHER PROOF DISCONNECT
B-1	BOILER	MCA 13	1	240	150	INTEGRAL	●			●	●				●	ELECTRICAL DIVISION WILL WIRE TO UNIT DISCONNECT
P-1 P-2	PUMP P-1,2	MCA 44	1	208	50	INTEGRAL									●	ELECTRICAL DIVISION WILL PROVIDE LOCAL DISCONNECT

STARTER SCHEDULE NOTES:

- SEE "WIRING & MOTORS" IN MECHANICAL GENERAL REQUIREMENTS FOR FULL DETAILS OF STARTERS, FUSES, OVERLOAD HEATERS, ETC.
- LABEL ON STARTERS FOR EQUIPMENT, FANS ETC., SHALL BE AS NOTED IN THEIR SCHEDULES. SEE SPECIFICATIONS FOR DETAIL.
- INTERLOCKS, FREEZE & SAFETY CONTROL WIRING SHALL BE BY MECHANICAL TRADE. SEE SPECIFICATIONS FOR DETAIL.
- STARTER LOCATION: – IN MECHANICAL ROOMS OR SERVICE ROOM STARTERS SHALL BE ON WALLS ADJACENT TO EQUIPMENT; – IN FINISHED ROOM STARTERS SHALL BE AT THE LIGHT SWITCH.
- ROOF EQUIPMENT THAT START FROM REMOTE PUSH BUTTON OR SELECTOR SWITCH SHALL HAVE THEIR STARTERS LOCATED IN THE NEAREST MECHANICAL ROOM TO THE EQUIPMENT.

ABBREVIATION:

- S.S. – SELECTOR SWITCH
P.B. – PUSH BUTTON
H/O/A – HAND-OFF-AUTOMATIC
P.L. – PILOT LIGHT
BMS – BUILDING MANAGEMENT SYSTEM

ELECTRICAL NOTES:

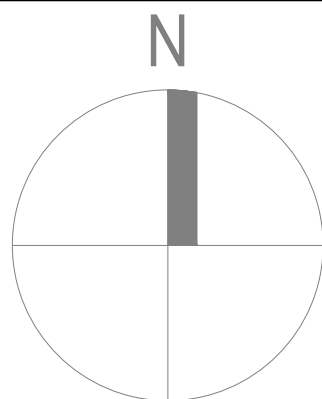
- ALL MOC P RATINGS ARE BASED ON MANUFACTURERS DATA, COORDINATE AND CONFIRM EXACT MOC P AND MCA WITH SHOP DRAWINGS AND MECHANICAL CONTRACTOR
- LABEL ON STARTERS FOR EQUIPMENT, FANS ETC., SHALL BE AS NOTED IN THEIR SCHEDULES. SEE SPECIFICATIONS FOR DETAIL.
- INTERLOCKS, FREEZE & SAFETY CONTROL WIRING SHALL BE BY MECHANICAL TRADE. SEE SPECIFICATIONS FOR DETAIL.
- STARTER LOCATION: – IN MECHANICAL ROOMS OR SERVICE ROOM STARTERS SHALL BE ON WALLS ADJACENT TO EQUIPMENT; – IN FINISHED ROOM STARTERS SHALL BE AT THE LIGHT SWITCH.
- ROOF EQUIPMENT THAT START FROM REMOTE PUSH BUTTON OR SELECTOR SWITCH SHALL HAVE THEIR STARTERS LOCATED IN THE NEAREST MECHANICAL ROOM TO THE EQUIPMENT.

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22-174



PROJECT NORTH

3	ADDENDUM EA-2	2026/10/07	DJ
3	ISSUED FOR PERMIT/TENDER	2025/09/25	DJ
2	ISSUED FOR COORDINATION	2025/08/18	DJ
1.	ISSUED FOR PERMIT	2023/09/15	DK
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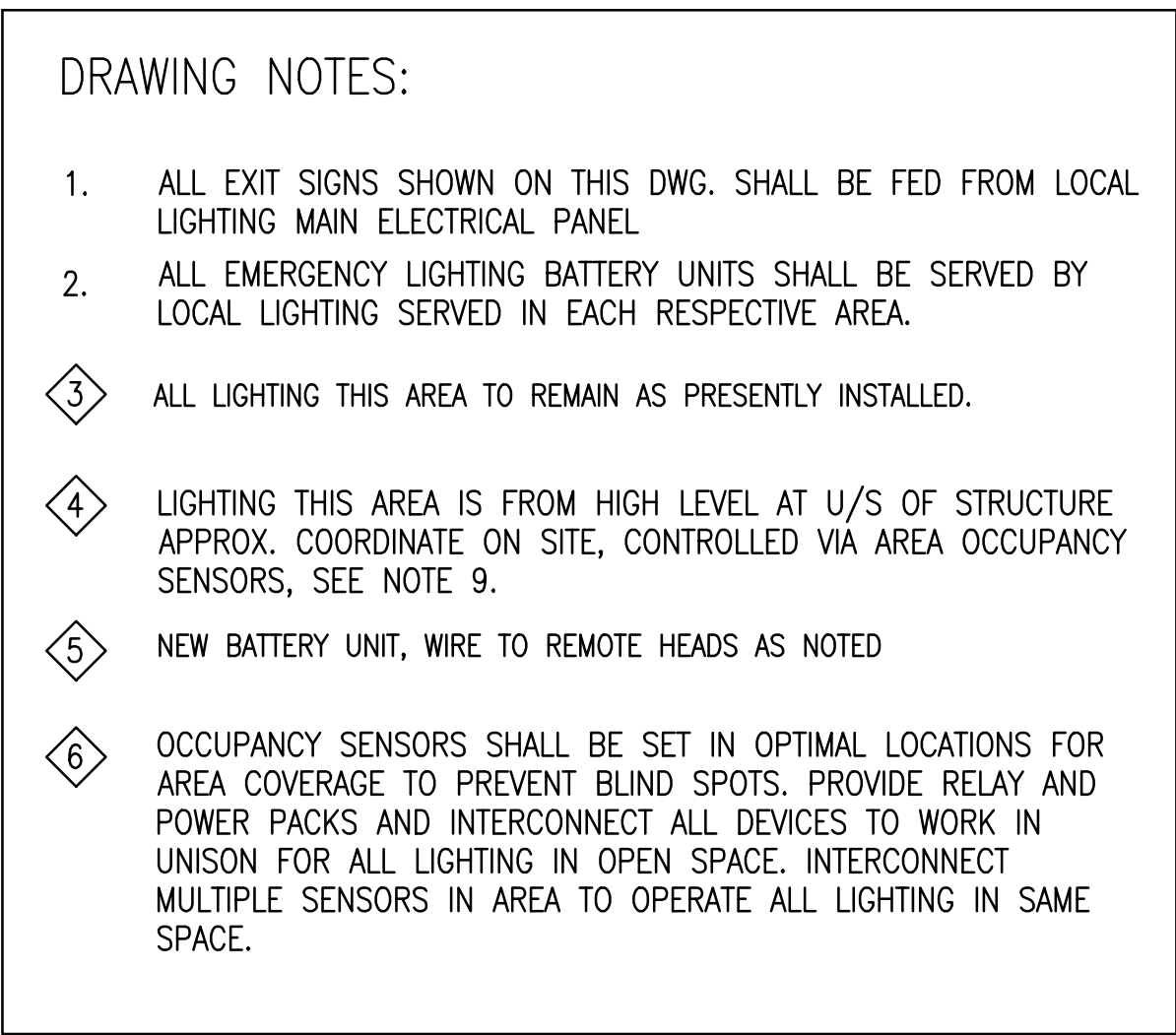
FLOOR PLANS :

SCHEDULES

SCALE : 3/16" = 1'-0"

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE
LOCATION : KOC DORM BLDG.

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1.	ISSUED FOR PERMIT		2023/09/15	DK	
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FLOOR PLANS :

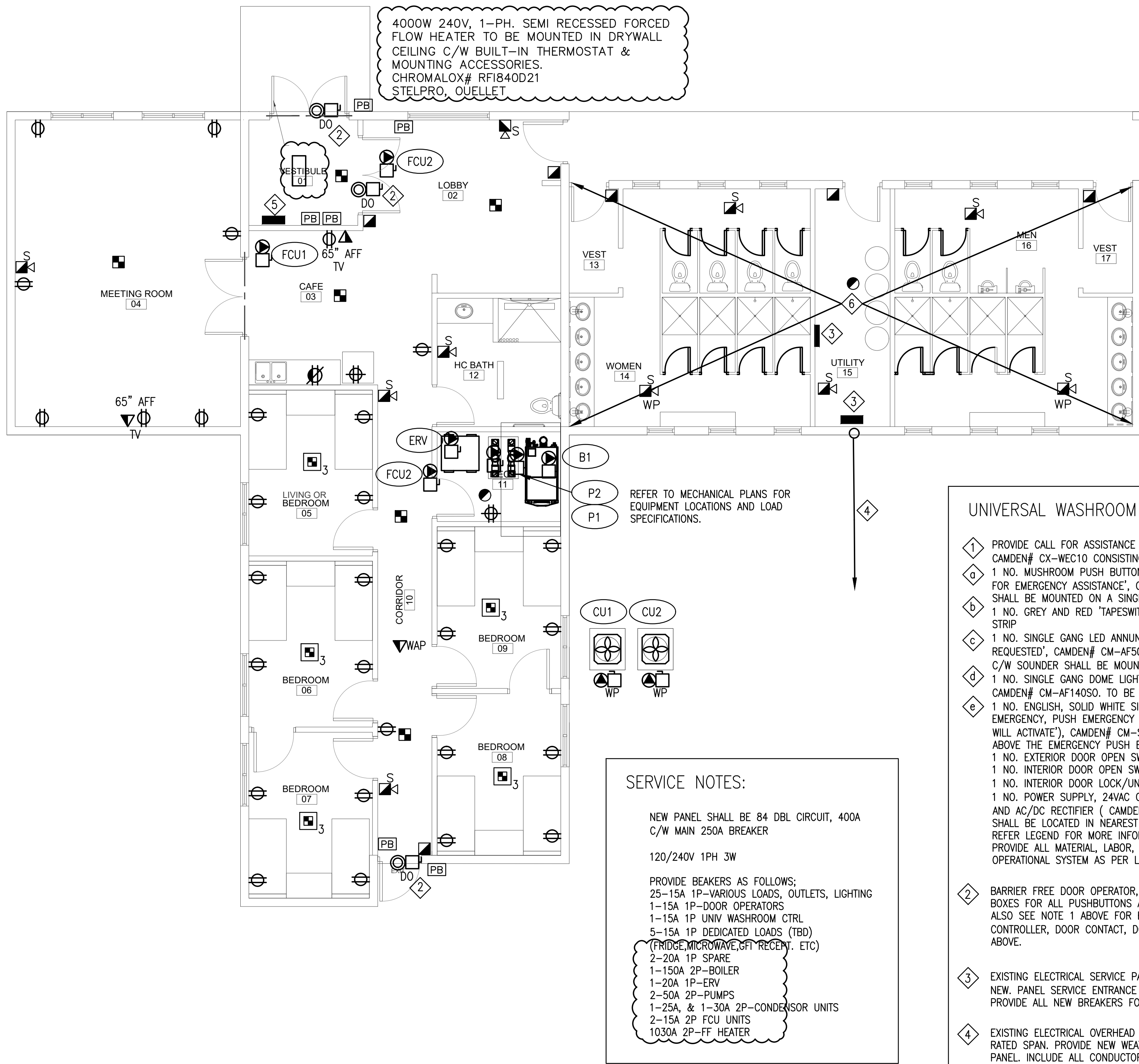
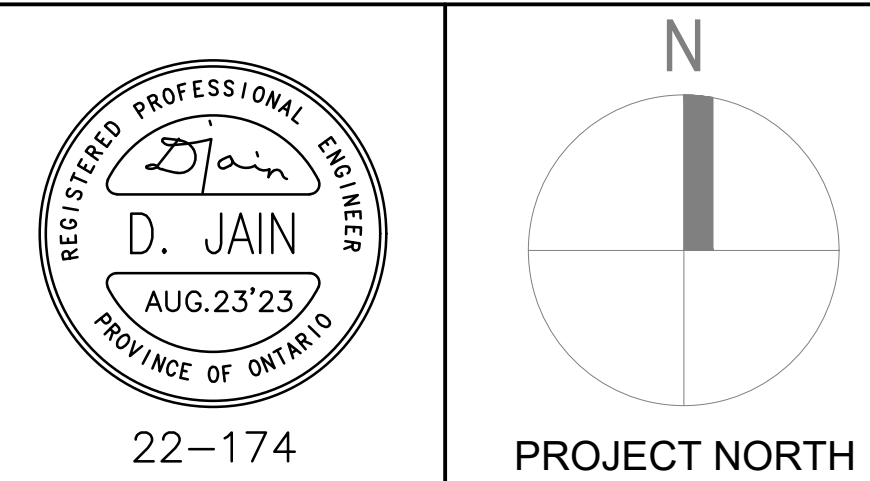
LIGHTING LAYOUT

SCALE : 3/16" = 1'-0"

PROJECT : 1766 QUEEN'S LINE ROAD
TOWNSHIP OF MINDEN HILLS
KINARK OUTDOOR
CENTRE

LOCATION : KOC DORM BLDG.

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		DWG. NO. E20



UNIVERSAL WASHROOM NOTES:

- PROVIDE CALL FOR ASSISTANCE SYSTEM IN WASHROOM WHICH SHALL BE CAMDEN# CX-WEC10 CONSISTING OF:
1 NO. MUSHROOM PUSH BUTTON, STAINLESS STEEL, PUSH/PULL, 'PRESS FOR EMERGENCY ASSISTANCE', CAMDEN# CM-450R/12. THE PUSH BUTTON SHALL BE MOUNTED ON A SINGLE GANG BACKBOX.
1 NO. GREY AND RED 'TAPESWITCH' 900mm LONG WATER PROOF PANIC STRIP
1 NO. SINGLE GANG LED ANNUNCIATOR C/W SOUNDER, 'ASSISTANCE REQUESTED', CAMDEN# CM-AF50150. THE SINGLE GANG LED ANNUNCIATOR C/W SOUNDER SHALL BE MOUNTED INSIDE THE B.F. WASHROOM.
1 NO. SINGLE GANG DOME LIGHT WITH SOUNDER, 'ASSISTANCE REQUESTED', CAMDEN# CM-AF140S0, TO BE MOUNTED OUTSIDE THE B.F. WASHROOM.
1 NO. ENGLISH, SOLID WHITE SIGN (152MMX270MM, 'IN THE EVENT OF AN EMERGENCY, PUSH EMERGENCY BUTTON AND AUDIBLE AND VISUAL SIGNAL WILL ACTIVATE', CAMDEN# CM-SE21A. THIS SIGN SHALL BE MOUNTED ABOVE THE EMERGENCY PUSH BUTTON OF CALL FOR ASSISTANCE SYSTEM.
1 NO. EXTERIOR DOOR OPEN SWITCH
1 NO. INTERIOR DOOR OPEN SWITCH
1 NO. INTERIOR DOOR LOCK/UNLOCK SWITCH
1 NO. POWER SUPPLY, 24VAC C/W 40VA STANDARD MOUNT TRANSFORMER AND AC/DC RECTIFIER (CAMDEN# CX-TRK24-50). THE POWER SUPPLY SHALL BE LOCATED IN NEAREST ACCESSIBLE CEILING SPACE. REFER LEGEND FOR MORE INFORMATION.
PROVIDE ALL MATERIAL, LABOR, CONDUITS AND WIRING FOR FULLY OPERATIONAL SYSTEM AS PER LATEST OBC REQUIREMENTS.
- BARRIER FREE DOOR OPERATOR, PROVIDE 120V POWER AND ROUGH IN BOXES FOR ALL PUSHBUTTONS AS DIRECTED BY HARDWARE INSTALLER. ALSO SEE NOTE 1 ABOVE FOR BARRIER FREE WASHROOM. CONNECT DOOR CONTROLLER, DOOR CONTACT, DOOR LATCH (LV CONTROL) TO CONTROLLER ABOVE.
- EXISTING ELECTRICAL SERVICE PANEL TO BE REMOVED AND REPLACED WITH NEW. PANEL SERVICE ENTRANCE RATED 400A 120/240V 1PH 3W, 42 CCT, PROVIDE ALL NEW BREAKERS FOR EXISTING LOADS.
- EXISTING ELECTRICAL OVERHEAD SERVICE TO BE UPGRADED TO NEW 400A RATED SPAN. PROVIDE NEW WEATHERHEAD AND SERVICE MAST TO NEW PANEL. INCLUDE ALL CONDUCTORS AND LEAVE TAIL FOR CONNECTION AT WEATHERHEAD.
-NEW OVERHEAD CONDUCTOR SPAN BY HYDRO ONE
-EXISTING POLE TOP TRANSFORMER TO BE UPGRADED AS REQUIRED BY HYDRO ONE.
-CONTRACTOR TO COORDINATE NEW SERVICE WITH HYDRO ONE (HONI)
- PROVIDE NEW SINGLE STAGE SINGLE ZONE FIRE ALARM CONTROL PANEL. MIRCOM FA10T, ULC/CSA LISTED. PROVIDE ALL DEVICES AND WIRING IN ACCORDANCE WITH OBC AND ULC STANDARDS. PROVIDE FIRE ALARM VERIFICATION TESTING AND REPORT UPON INSTALLATION COMPLETION.
- EXISTING ELECTRICAL INSTALLATIONS THIS AREA TO REMAIN AS PRESENT;Y INSTALLED

SERVICE NOTES:

NEW PANEL SHALL BE 84 DBL CIRCUIT, 400A
C/W MAIN 250A BREAKER

120/240V 1PH 3W

PROVIDE BEAKERS AS FOLLOWS;
25-15A 1P-VARIOUS LOADS, OUTLETS, LIGHTING
1-15A 1P-DOOR OPERATORS
1-15A 1P UNIV WASHROOM CTRL
5-15A 1P DEDICATED LOADS (TBD)
(Fridge, MICROWAVE, ST RECENT, ETC)
2-20A 1P SPARE
1-150A 2P-BOILER
1-20A 1P-ERV
2-50A 2P-PUMPS
1-25A, & 1-30A 2P-CONDENSOR UNITS
2-15A 2P FCU UNITS
1030A 2P-FF HEATER

FLOOR PLANS :
POWER LAYOUT

SCALE : 3/16" = 1'-0"

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CENTRE
LOCATION : KOC DORM BLDG.

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